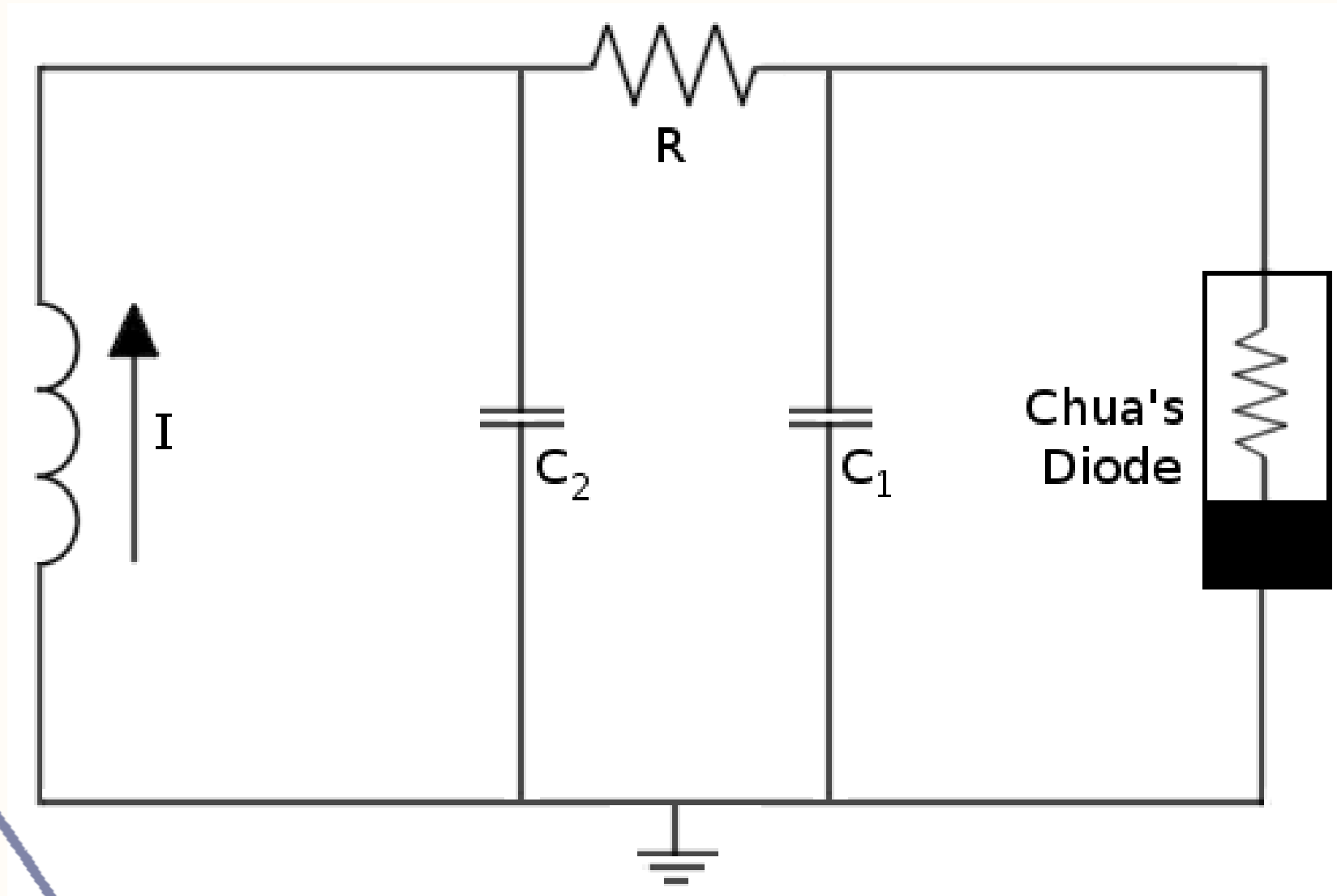


Observation du caractère chaotique du circuit de Chua

Introduction

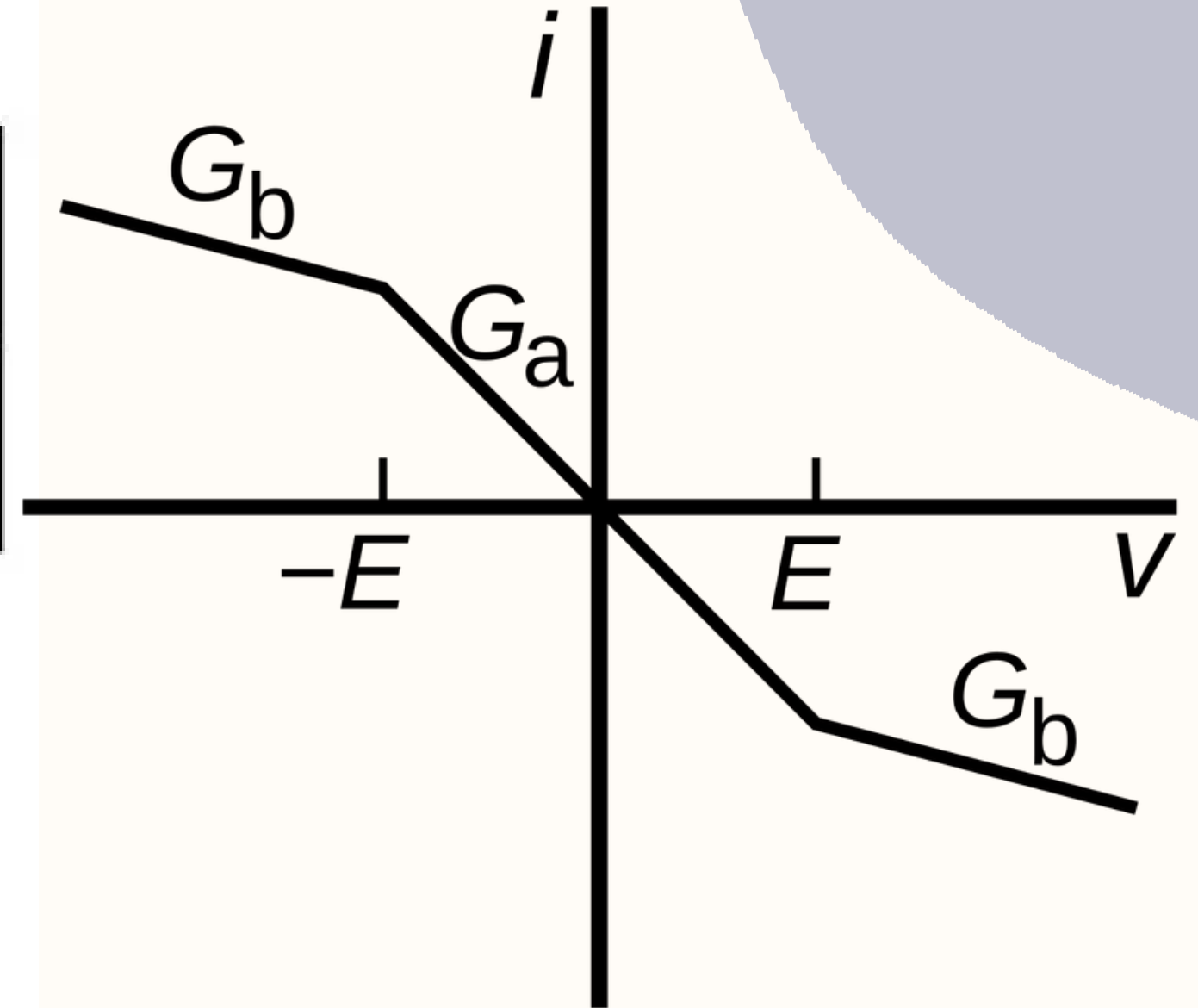
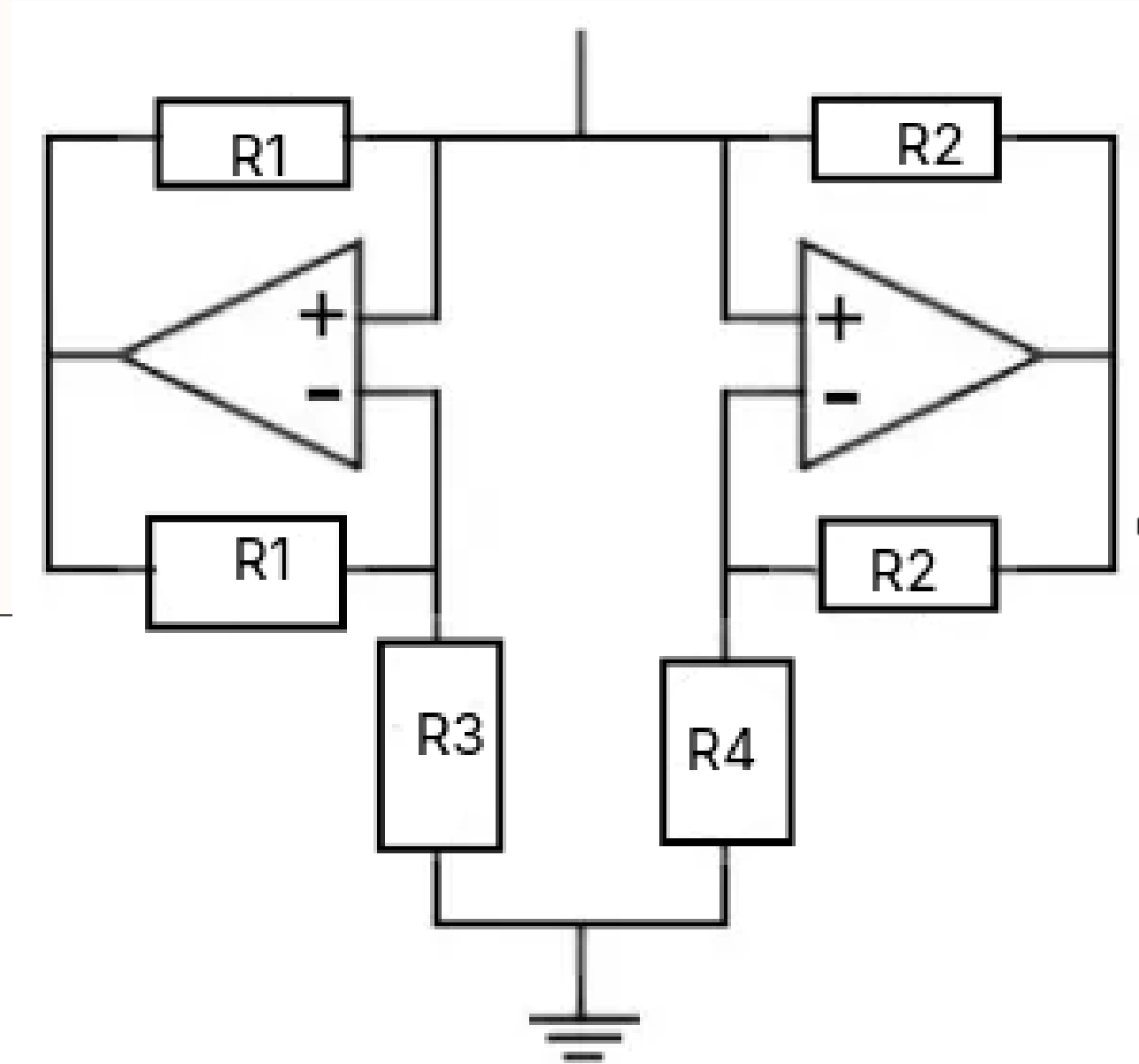




Le circuit de Chua

$$\begin{aligned} C_1 \frac{dv_1}{dt} &= \frac{v_2 - v_1}{R} - g(v_1), \\ C_2 \frac{dv_2}{dt} &= \frac{v_1 - v_2}{R} + i_L, \\ L \frac{di_L}{dt} &= -v_2. \end{aligned}$$

La Diode de Chua



$$G_b = -\frac{1}{R_3} - \frac{1}{R_4}$$

$$G_a = -\frac{1}{R_4} + \frac{1}{R_1}$$

$$E = \frac{R_3}{R_3 + R_1} E_{sat}$$

$$g(V) = G_b V + \frac{1}{2} (G_a - G_b) (|V + E| - |V - E|)$$

$$x_{k+1} = r * f(x_k)$$

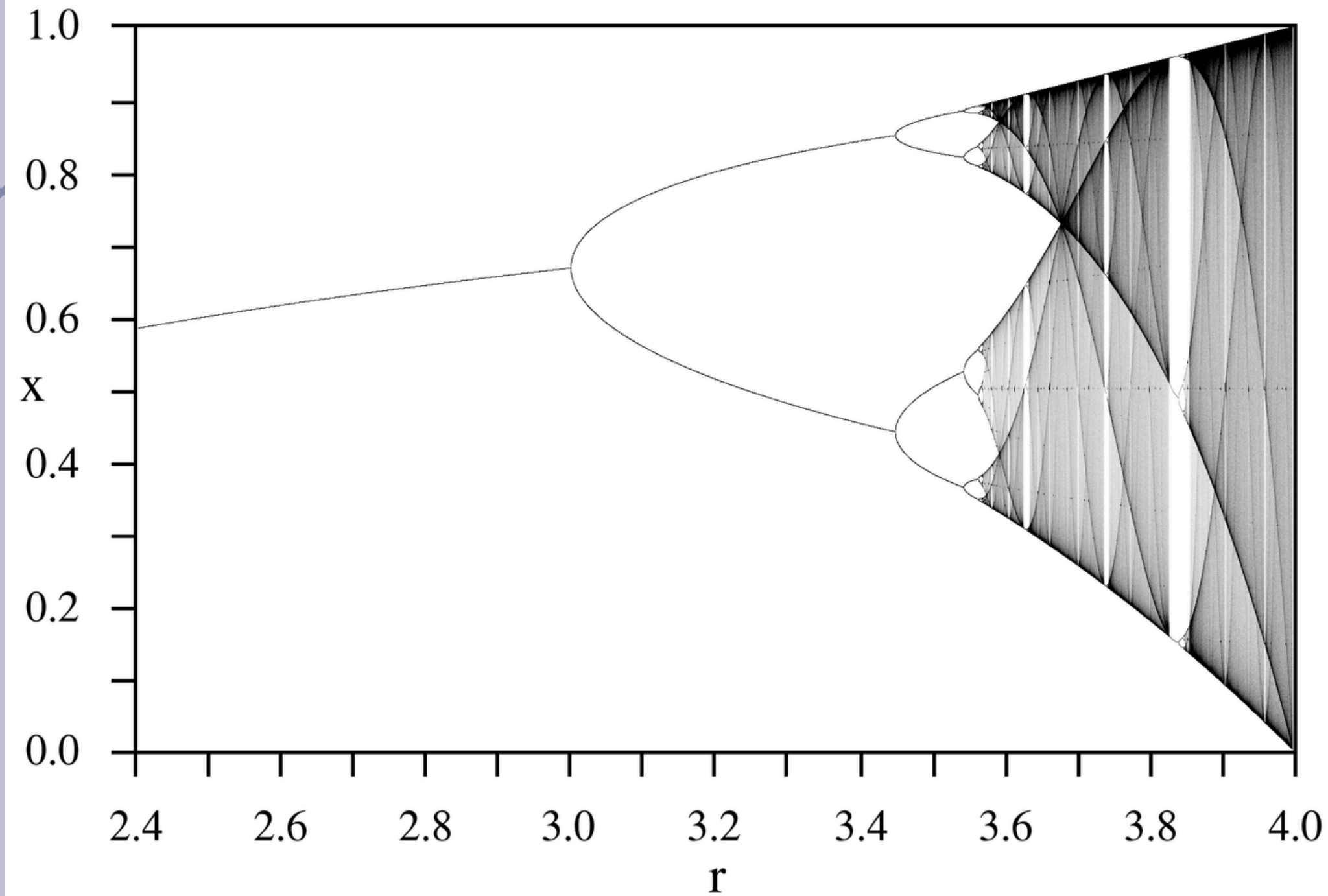
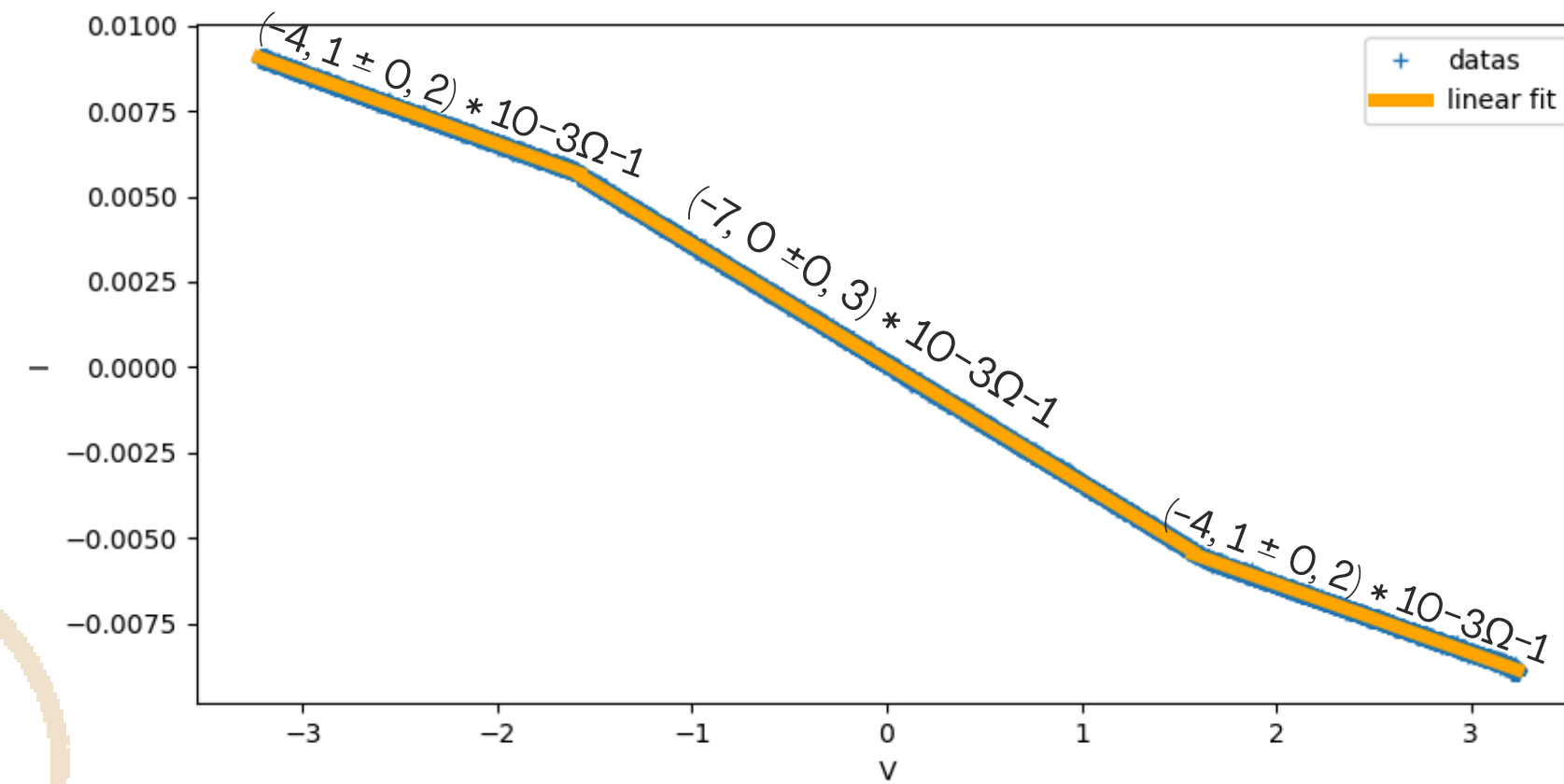


Diagramme de
bifurcation

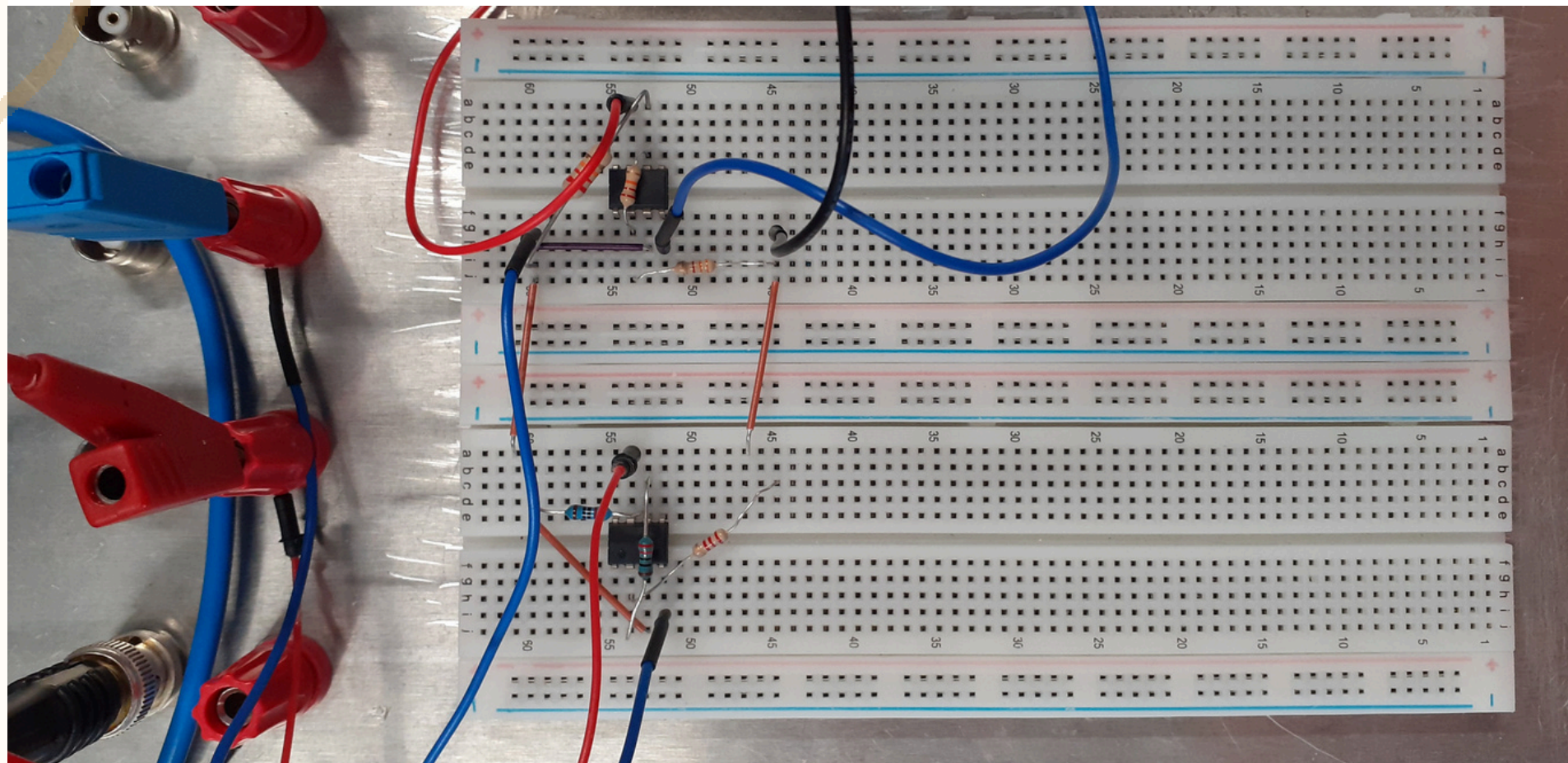
$$\delta_n = \frac{R_{n+1} - R_n}{R_{n+2} - R_{n+1}}$$

$$\delta = \lim_{n \rightarrow \infty} \delta_n \approx 4,669$$

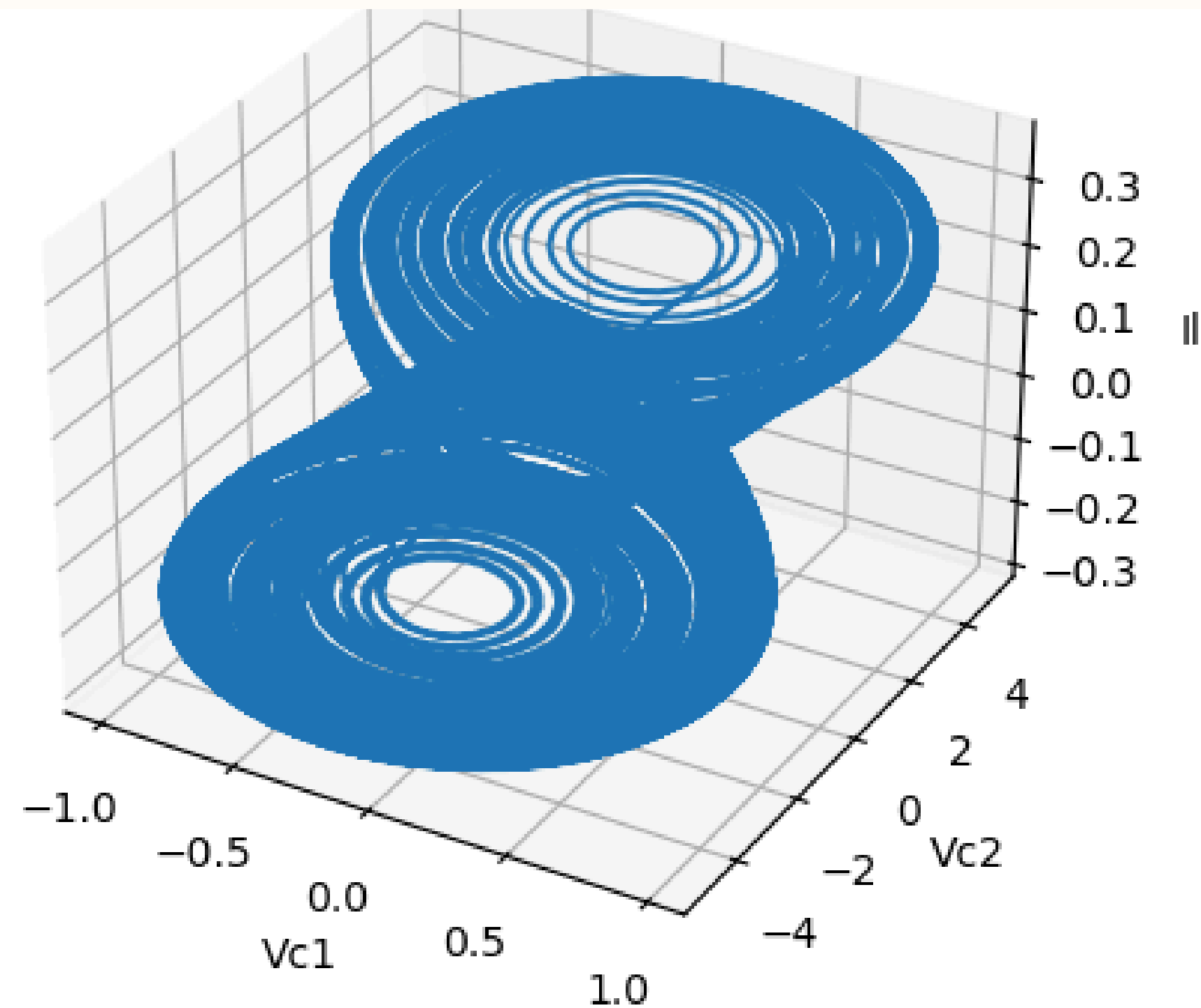
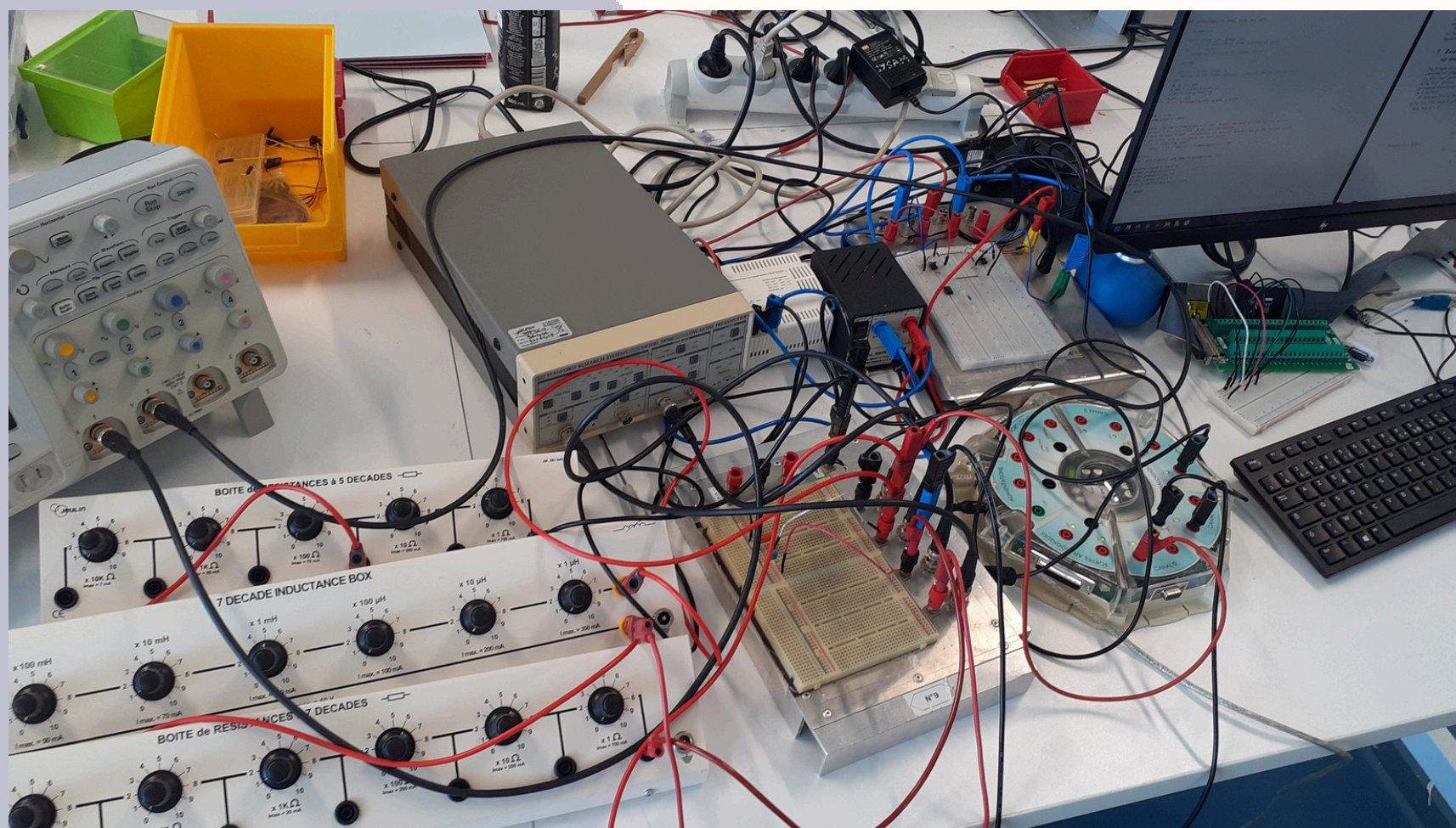
La constante de
Feigenbaum



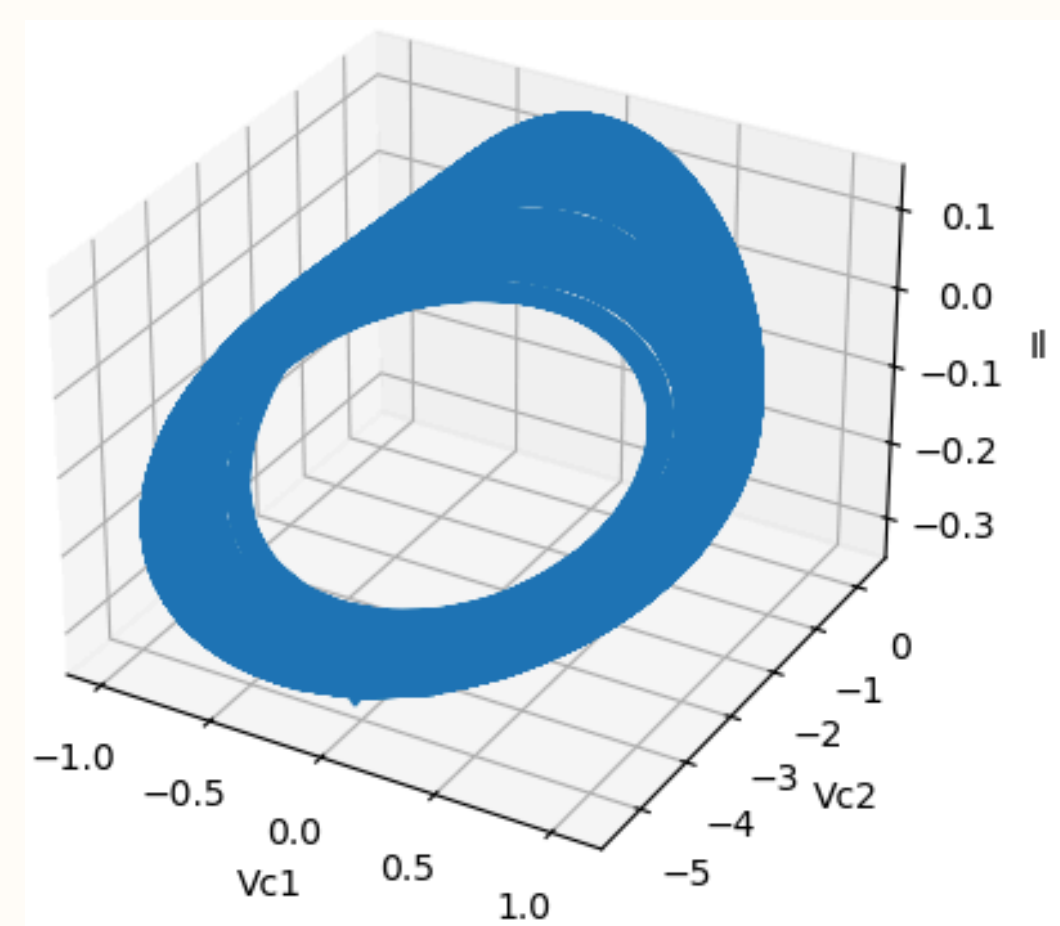
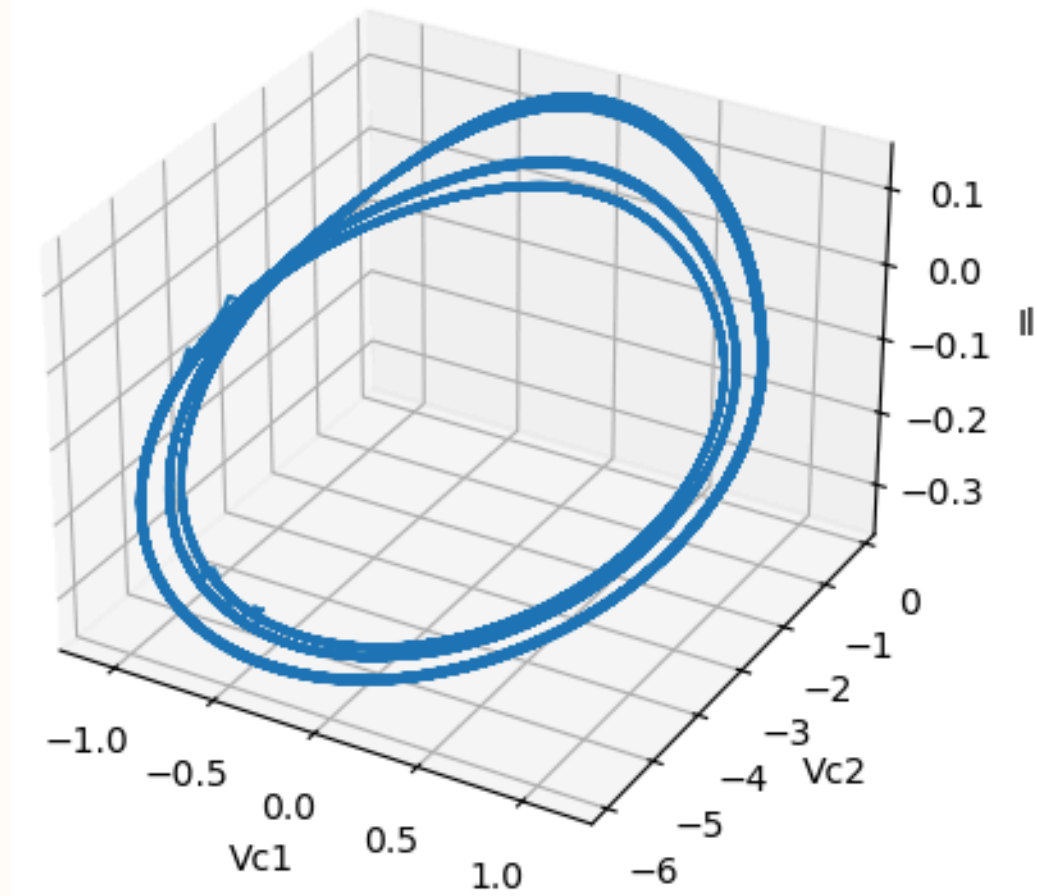
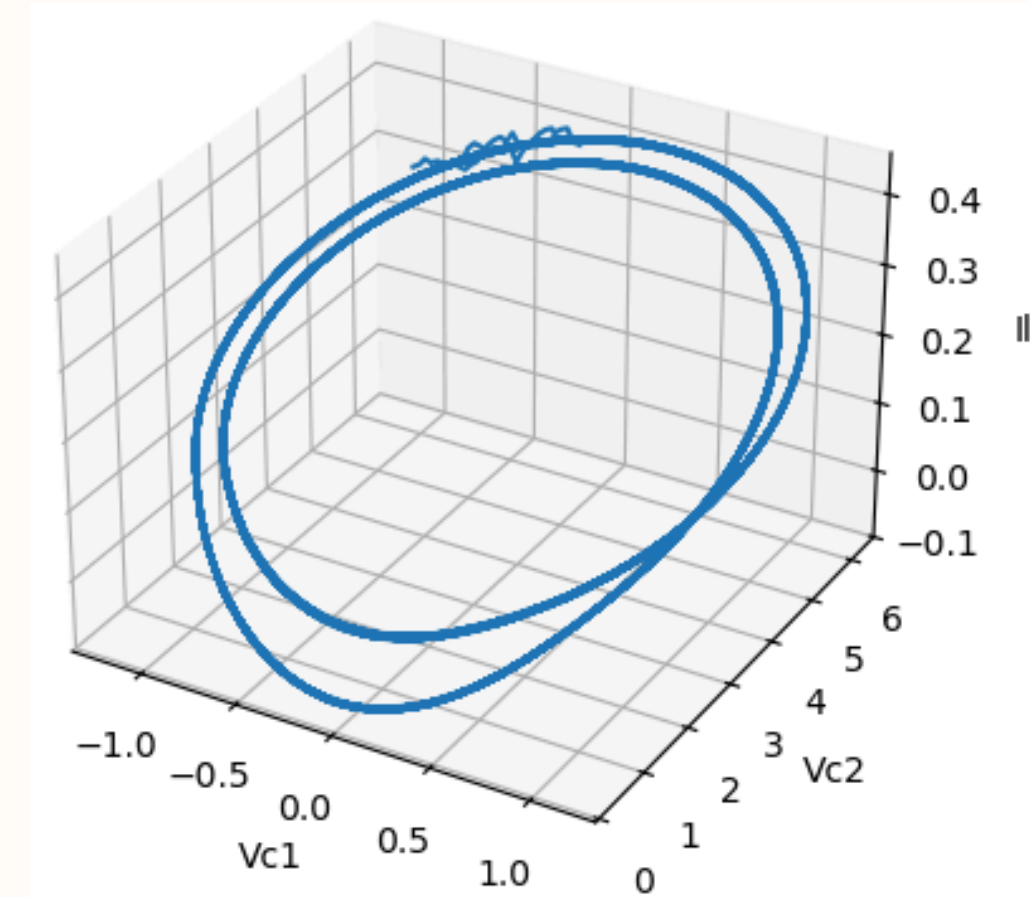
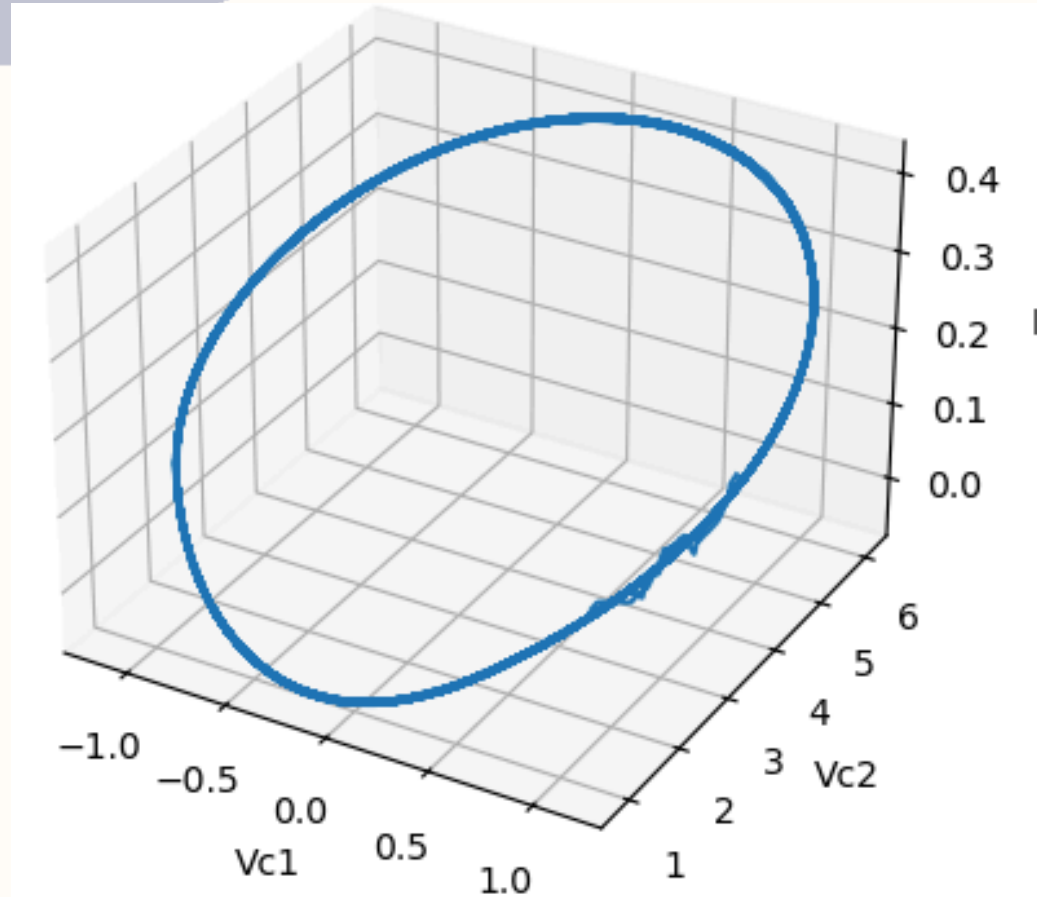
Réalisation de la diode



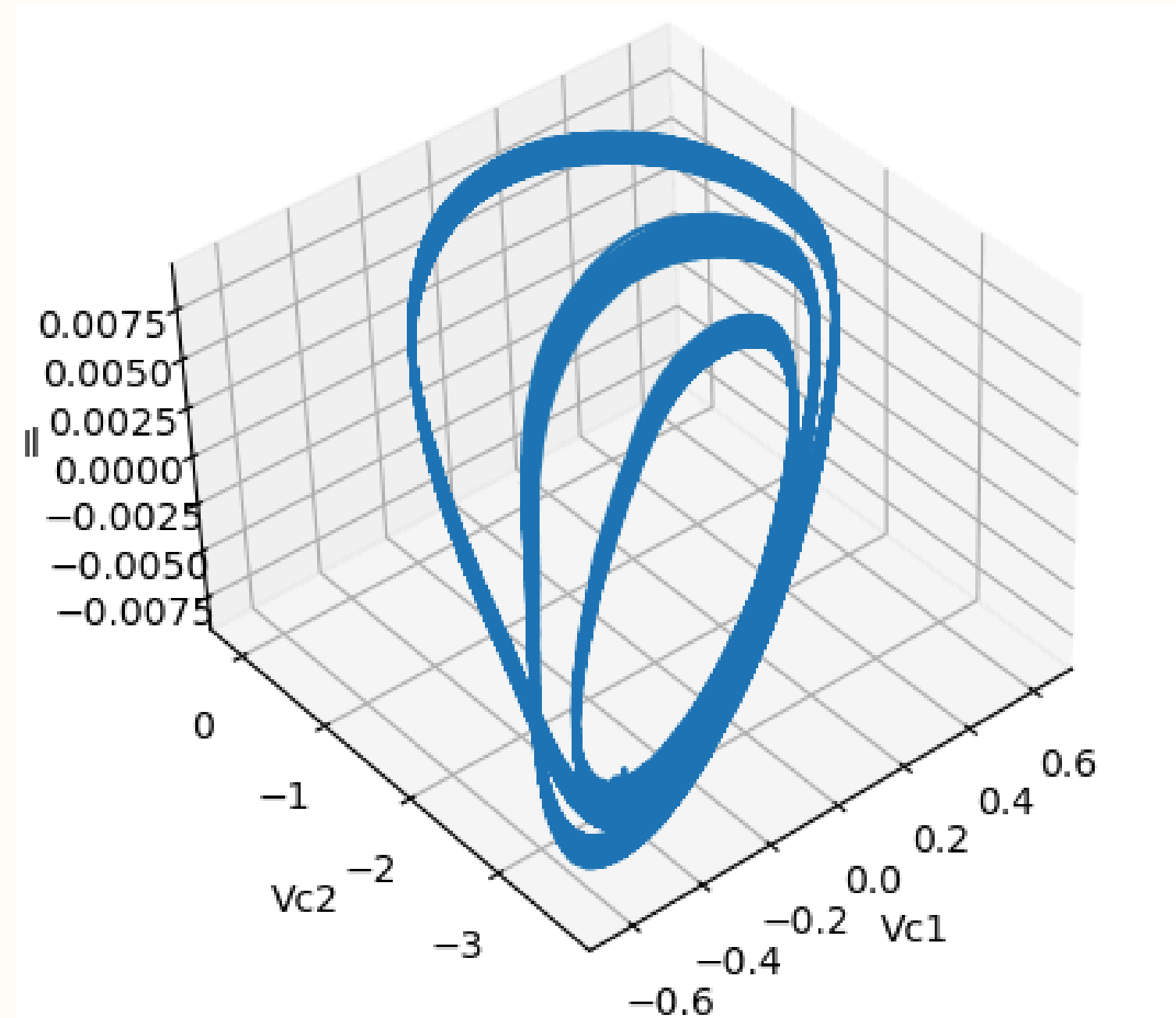
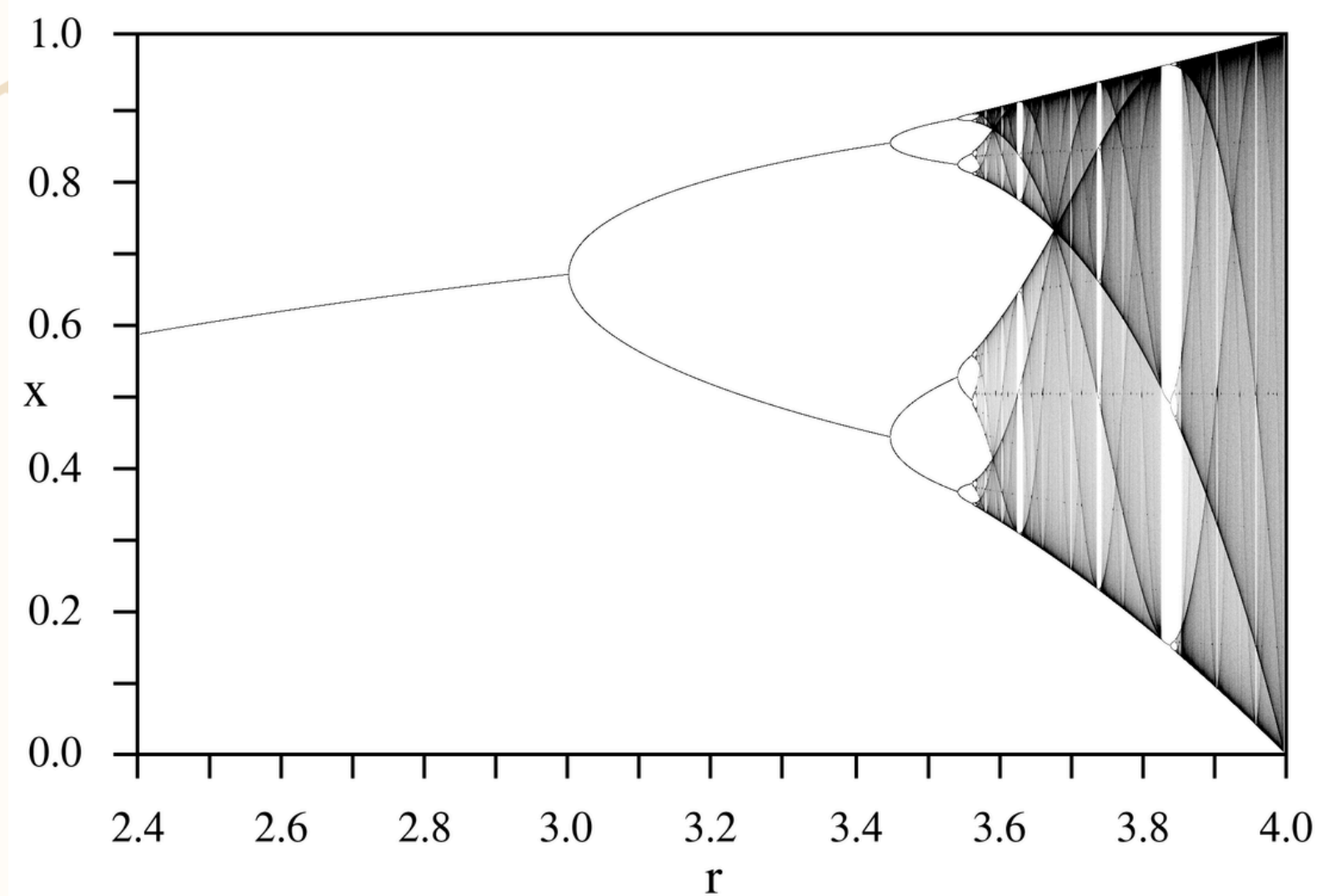
réalisation du circuit



Différents régimes



Différents régimes



Mesure de R aux transitions de régimes

	A	B	C	D	E	F
1	L(mH)	R0->1(Ohm)	R1->2(Ohm)	R2->4(Ohm)	R4->8(Ohm)	R8->16(Ohm)
2	14,95	1716	1600	1561	1555	1553
3	15	1715	1590	1552	1543	1541
4	15,05	1684	1558	1516	1504	1501
5	15,1	1676	1544	1499	1490	1487
6	15,2	1636	1492	1437	1428	1424
7	15,3	1594	1430	1359	1338	1334
8	15,4	1594	1435	1372	1351	1347

Une valeur
expérimentale de
la constante de
Feigenbaum ?

Delta1	Delta2	Delta3
2,80700236	4,53061224	3,89285714

u(Delta1)	u(Delta2)	u(Delta3)
0,22962405	0,62451771	0,95045914

Valeur théorique :

$$\delta \approx 4.669$$



Merci de nous
avoir écouté